Individual Coursework Submission Form

Specialist Masters Programme

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Mortgage Repayment Analysis

A detailed analysis of mortgage Instalments and outstanding Balances

By

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1. (Question 1) Mortgage Repayment Calculations

A borrower takes out a mortgage of £250,000.00 to purchase a new house. This amount is paid in level instalments in arrears for next 25 years. Assuming the effective rate of interest 8.25% per annum.

(i) Calculate the Level Annual Instalment

The level annual installment R, we use the following formula

$$R = \frac{C}{a_{\overline{n}|}}$$

here,

- C = £250,000 (Mortgage amount)
- $a_{\overline{n}|}$= the annuity factor for n years at the interest rate i
- i = 8.25% per annum = 0.0825 (effective interest rate)
- n = 25 years (loan term)

The annuity factor $a_{\overline{n}|}$ is calculated as follows:

$$a_{\overline{n}|} = \frac{1-(1+i)^{-n}}{i} = \frac{1-(1+0.0825)^{-25}}{0.0825} = \frac{1-0.1378169}{0.0825} = 10.450705$$

Now that we have the annuity factor, we can calculate the level annual instalment R:

$$R = \frac{250000}{10.450705} = 23921.832$$

Thus, the level annual instalment is approximately **£23921.83**.

(ii) Mortgage Outstanding at Time t=1 (One Year)

Next, we calculate the mortgage outstanding immediately after the first payment, i.e., at time t=1 year. The mortgage outstanding at time t, denoted L_1, is given by the following formula:

$$L_t = C(1+i) - R$$

here,

- C= £250000 (mortgage)
- i = 0.0825 (effective rate of interest)
- R= 23921.83 (Level annual installment)

Substitute the values into the formula:

L_1= 250,000(1+0.0825) - 23921.83

L_1= 246703.17

Thus, the mortgage outstanding after the first year is **£246703.17**.

(iii) Constructing an Excel Spreadsheet to Calculate Mortgage Outstanding Over 25 Years

To calculate the mortgage outstanding at each year t, we can use the following iterative formula:

$$L_t = L_{t-1}(1 + i) - R$$

Here,

- L_t = the mortgage outstanding at year t
- L_{t-1} = the mortgage outstanding at the previous year
- i = the effective interest rate (8.25%)
- R = the annual level payment (£23,922.53)

As an **example**, to calculate the mortgage outstanding at time t =2, use the following:

$L_2 = L_1\ (1 + 0.0825)\ -\ 23921.83$ = 246703.17 (1.0825) - 23921.83 = 243134.35

So, the mortgage outstanding after the second year is approximately **£243,134.35**.

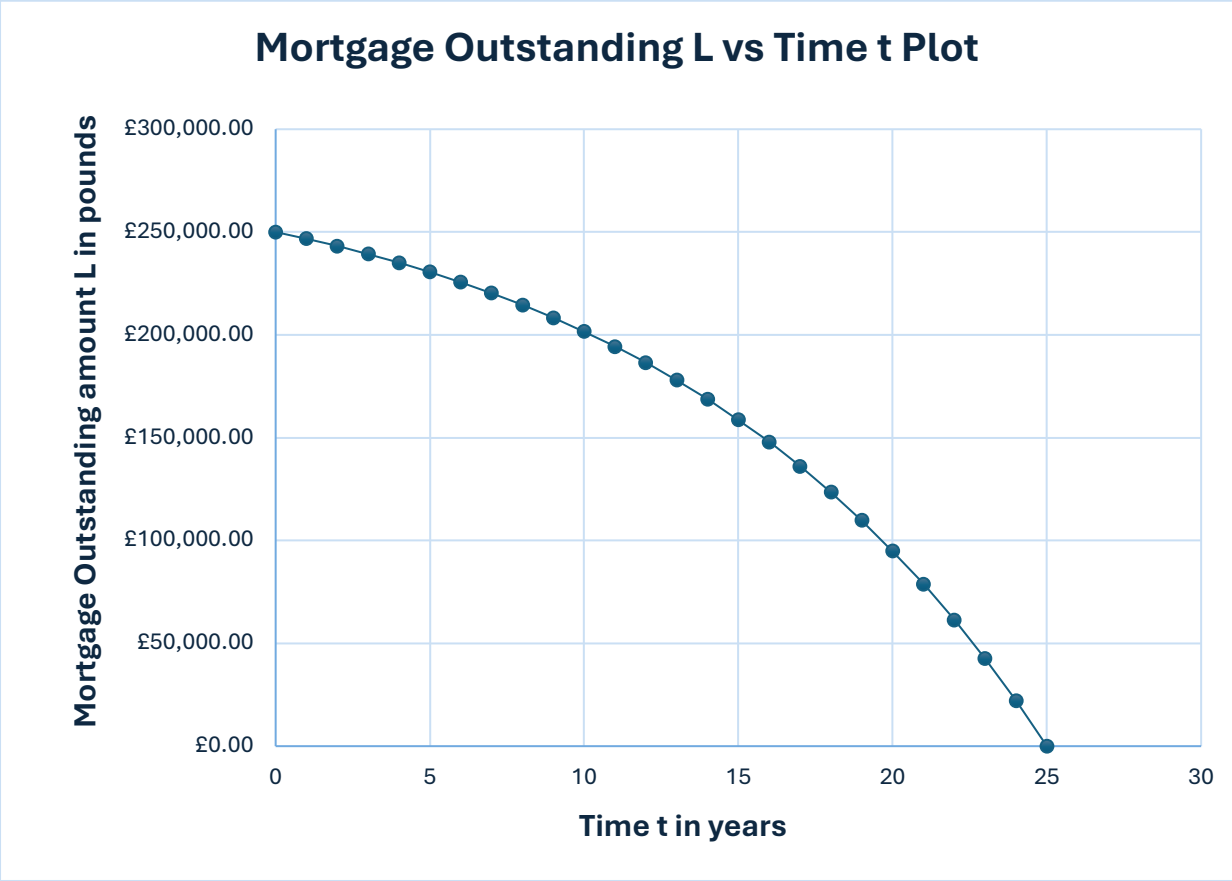
Continue this process for each year from t=1 to t=25, and you will have the mortgage outstanding for all 25 years.

The Calculated table for Mortgage Outstanding L_t is given below:

Time, t	Loan Outstanding,L
0	£250,000.00
1	£246,703.17
2	£243,134.35
3	£239,271.10
4	£235,089.13
5	£230,562.15
6	£225,661.70
7	£220,356.96
8	£214,614.58
9	£208,398.45
10	£201,669.49
11	£194,385.39
12	£186,500.35
13	£177,964.80
14	£168,725.06
15	£158,723.05
16	£147,895.86
17	£136,175.44
18	£123,488.08
19	£109,754.02
20	£94,886.89
21	£78,793.23
22	£61,371.84
23	£42,513.18
24	£22,098.69
25	£0.00

(iv) Use Excel to Plot a Graph of the Mortgage Outstanding at Time t, L_t, Against Time t

With the help of Excel, we plot the following graph which shows Mortgage outstanding amount in Y-axis and time in X-axis.



Interpretation of the plot:

The plot of the mortgage outstanding clearly shows a **decreasing curve** which highlights the dynamic shift between interest and principal payments. In the early years, the outstanding balance decreases slowly as most of the monthly payment is allocated to interest due to the large initial mortgage amount. This result in flatter curve during this face. as the loan progresses, the interest portion of each payment diminishes, allowing a large share to reduce the principal. this accelerates the rate if balance reduction, creating a steeper curve in the middle years. Toward the end of the term, the remaining balance is small, and while most of the payment goes toward the principal, the absolute reduction slows, causing the curve to flatten. This pattern reflects the natural progression of fixed-rate loans, balancing interest and principal over time to achieve full repayment.

(v) Calculate the Total Amount of Interest Payable Over the Term of the Mortgage.

To calculate the total interest paid over the term of the mortgage, we follow two steps:
We need to find the Total Amount Paid Over the 25-Year Term:

$$\text{Total Paid} = R \times n$$
$$= 23921.83 \times 25$$
$$= 598045.80$$

Then we can find the Total Interest Paid: C = 250,000 and Total paid = 598045.80

$$\text{Total Interest Paid} = \text{Total amount paid} - C$$
$$= 598045.80 - 250000$$
$$= 348045.80$$

The total amount interest payable over the term of the mortgage is **£348045.80**.

(vi) Calculate the Amount to Be Repaid After 12 Years and 7 Months

Suppose the borrower wins the lottery and decides to pay off the mortgage in full after 12 years and 7 months. To calculate the amount to be repaid at that time, we proceed as follows:

Determine the Outstanding Balance at Year 12 and 7 months:

$$L_{12.58} = L_{12} + (L_{13} - L_{12})(7/12)$$

Here,

- L_{12} = £186,500.35
- L_{13} = £177,964.80

Substituting these values we get,

$$L_{12.58} = 186{,}500.35 + (177{,}964.80 - 186{,}500.35)*(7/12)$$
$$L_{12.58} = £181{,}521.28$$

Thus, the Amount to Be Repaid After 12 Years and 7 Months is £**181,521.28**.

2. (Question 2) Repetition with Different Interest Rates

(i): Level Annual Instalments at Different Interest Rates

We repeat the question 1(i) calculations for the different interest rates.
The level annual installment R, we use the following formula

$$R = \frac{C}{a_{\overline{n}|}}$$

here,

- C = £250,000 (Mortgage amount)
- $a_{\overline{n}|}$= the annuity factor for n years at the interest rate i

a) i = 1.25%

The annuity factor $a_{\overline{n}|}$ is calculated as follows:

$$a_{\overline{n}|} = \frac{1-(1+i)^{-n}}{i} = \frac{1-(1+0.0125)^{-25}}{0.0125} = \frac{1-0.7330341}{0.0125} = 21.357269$$

Now that we have the annuity factor, we can calculate the level annual instalment R:

$$R = \frac{250000}{21.357269} = 11{,}705.62$$

Thus, the level annual instalment is approximately **£11,705.62**

b) i = 8.25%

The annuity factor $a_{\overline{n}|}$ is calculated as follows:

$$a_{\overline{n}|} = \frac{1-(1+i)^{-n}}{i} = \frac{1-(1+0.0825)^{-25}}{0.0825} = \frac{1-0.1378169}{0.0825} = 10.450705$$

Now that we have the annuity factor, we can calculate the level annual instalment R:

$$R = \frac{250000}{10.450705} = 23921.832$$

Thus, the level annual instalment is approximately **£23921.83**.

c) i = 15.25%

The annuity factor $a_{\overline{n}|}$ is calculated as follows:

$$a_{\overline{n}|} = \frac{1-(1+i)^{-n}}{i} = \frac{1-(1+0.1525)^{-25}}{0.1525} = \frac{1-0.0287724}{0.1525} = 6.3687053$$

Now that we have the annuity factor, we can calculate the level annual instalment R:

$$R = \frac{250000}{6.3687053} = 39{,}254.45$$

Thus, the level annual instalment is approximately **£39,254.45**.

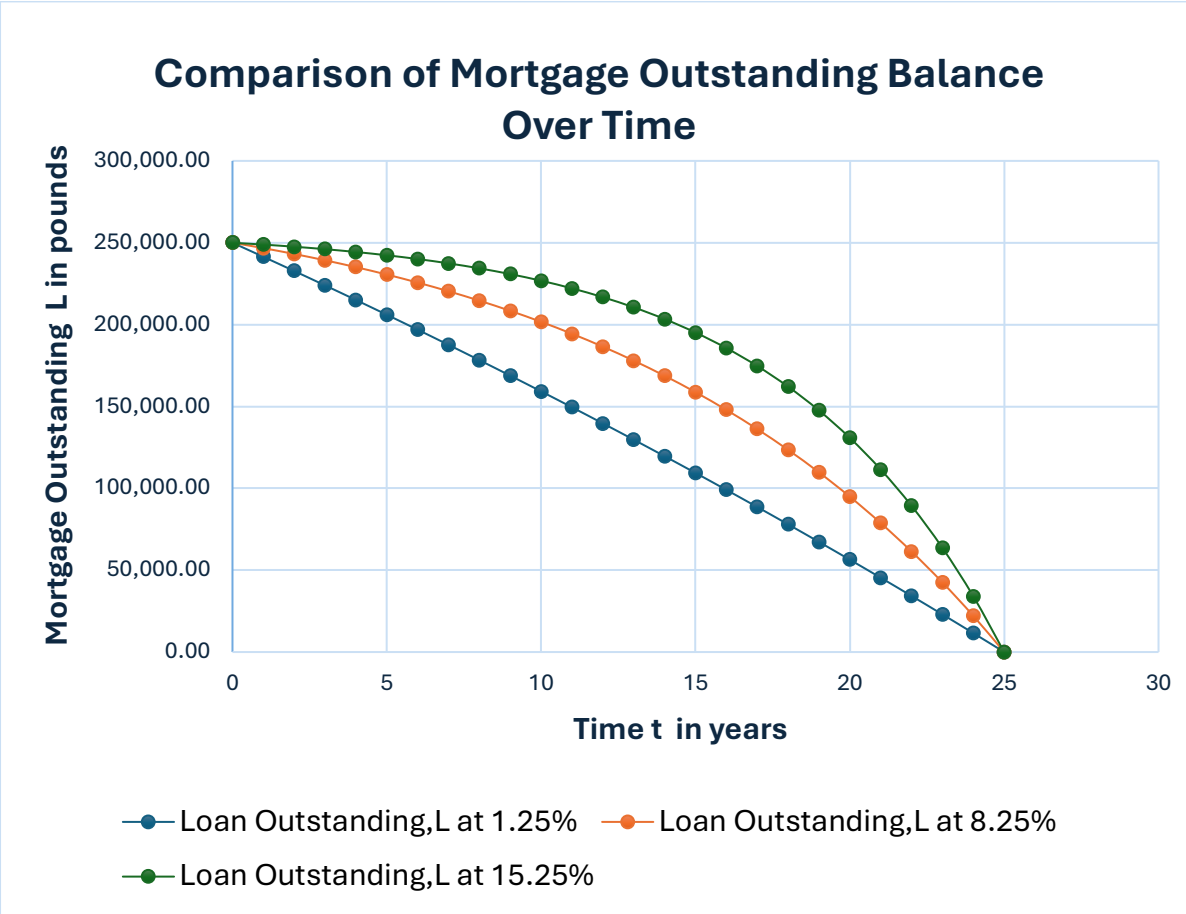
ii) Constructing an Excel Spreadsheet to Calculate Mortgage Outstanding Over 25 Years for different rate of interest i.e. 1.25%, 8.25% and 15.25%

The following table shows the mortgage outstanding for different rate of interest

Time, t	Loan Outstanding,L		
	at 1.25%	at 8.25%	at 15.25%
0	250,000.00	250,000.00	250,000.00
1	241,419.38	246,703.17	248,870.55
2	232,731.51	243,134.35	247,568.87
3	223,935.04	239,271.10	246,068.67
4	215,028.61	235,089.13	244,339.70
5	206,010.85	230,562.15	242,347.06
6	196,880.37	225,661.70	240,050.54
7	187,635.76	220,356.96	237,403.80
8	178,275.59	214,614.58	234,353.43
9	168,798.41	208,398.45	230,837.89
10	159,202.78	201,669.49	226,786.22
11	149,487.20	194,385.39	222,116.67
12	139,650.17	186,500.35	216,735.01
13	129,690.18	177,964.80	210,532.66
14	119,605.69	168,725.06	203,384.44
15	109,395.14	158,723.05	195,146.12
16	99,056.97	147,895.86	185,651.46
17	88,589.56	136,175.44	174,708.86
18	77,991.31	123,488.08	162,097.52
19	67,260.59	109,754.02	147,562.95
20	56,395.73	94,886.89	130,811.85
21	45,395.06	78,793.23	111,506.21
22	34,256.88	61,371.84	89,256.46
23	22,979.48	42,513.18	63,613.63
24	11,561.10	22,098.69	34,060.26
25	0.00	0.00	0.00

(iii) Use Excel to Plot a Graph of the Mortgage Outstanding at Time t, L_t, Against Time t for three different interest rates

The graph compares the mortgage outstanding balance over time for three different loan interest rates: 1.25%, 8.25%, and 15.25%. The x axis represents time (in years), while the y-axis represents the mortgage outstanding balance.



Interpretation and key observation:

The graph clearly shows how the outstanding balance decreases over a 25-year period for each interest rate. The graph, we can clearly see the rate of decrease is faster for lower rates. Higher interest rates result in slower payment of the principal balance, as more of the payment goes towards interest rather than reducing the outstanding balance. The slope analysis for different interest rates:

- **1.25% Interest rate (Blue Curve)**: The curve is the **steepest**, which indicates a faster reduction in outstanding balance due to less interest payment burden. The borrower benefits from lower interest, with a high potential to repay and reduce interest's cost. In lenders prospective its doesn't seem beneficial due to lower overall interest income and the possibility of early payoff.

- **8.25% Interest rate (Orange Curve)**: The curve is **less steep than the 1.25% curve**. The slope remains moderate throughout. The borrower has moderate level payments but slower equity buildup and has reduced opportunity for early payoff. The lender earns more interest income than at lower rates but faces moderate prepayment risks.

- **15.25% Interest rate (Green Curve)**: The curve is **less steep**, especially in the early years, indicating the slowest reduction in the outstanding amount. The slope increases slightly towards the end, as more of the repayment starts going to the principal. The borrower faces high payments, slower principal reduction and very limited opportunity for prepayments. The lenders earn significant interest income with lower prepayment risk.